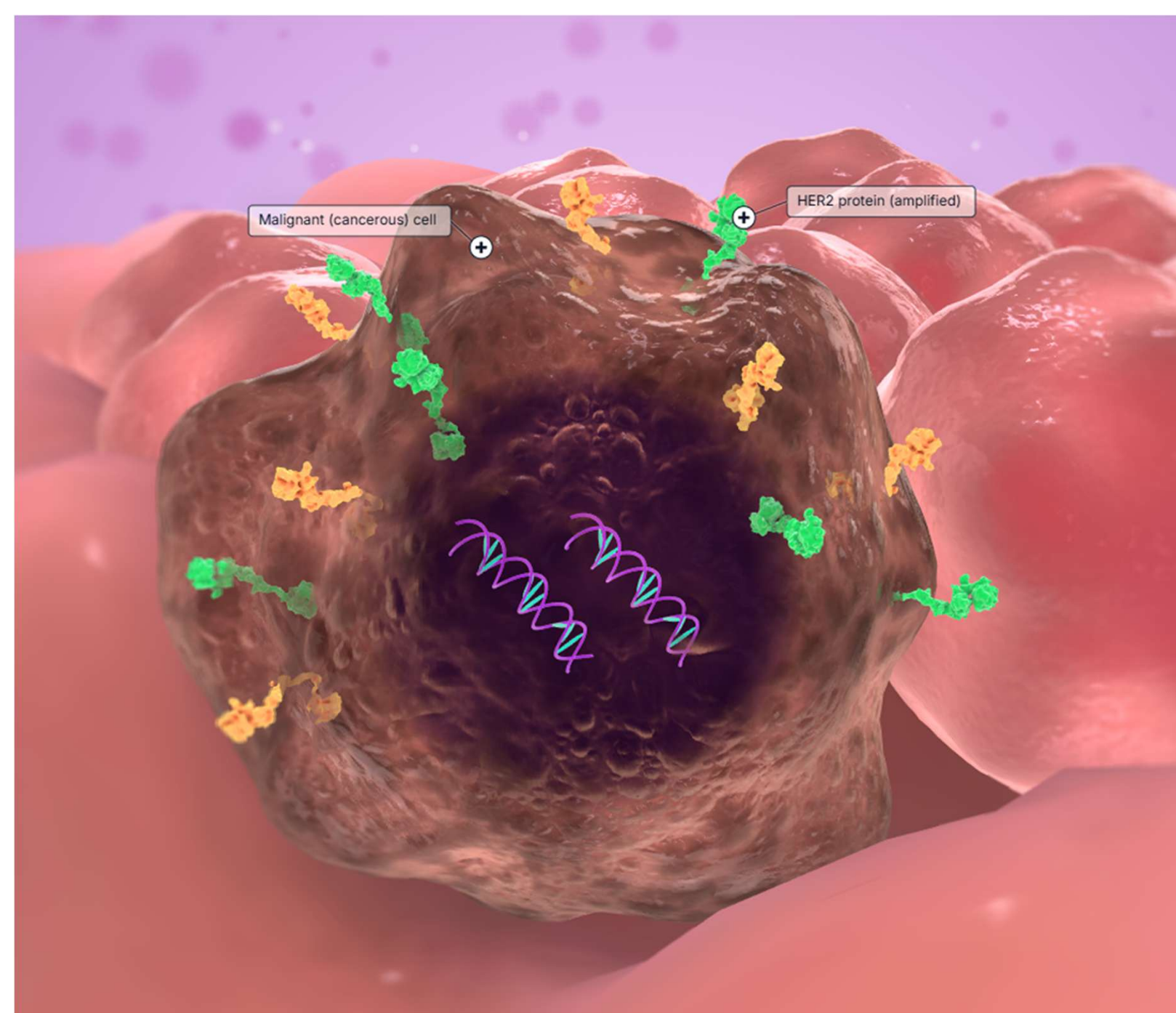
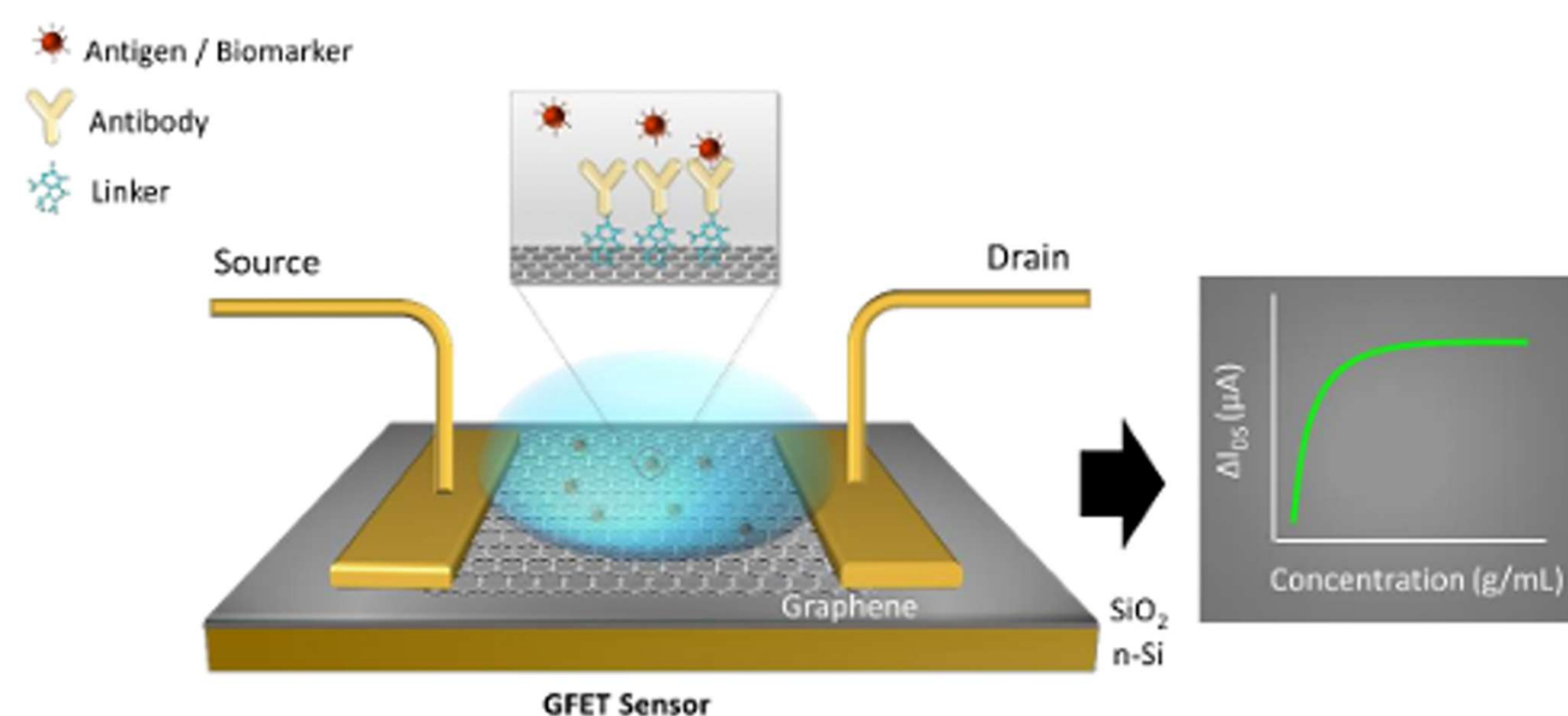




Problem and Biological Motivation

- HER2 is a clinically important breast cancer biomarker.
- Tumor-derived molecules can move through lymphatic pathways toward regional lymph nodes.
- A sensor cannot detect HER2 until molecules reach the sensing surface.
- Therefore, transport, sensor placement, and local exposure are part of the detection problem.

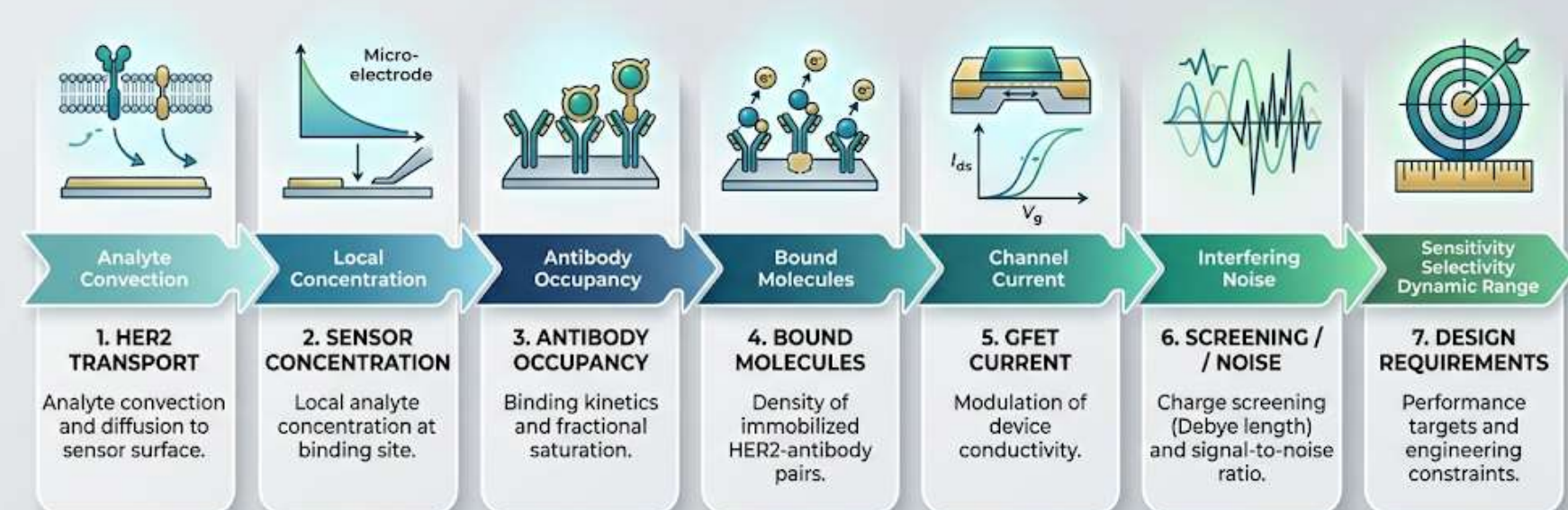


GFET-Based HER2 Detection Principle

- Antibodies immobilized on graphene selectively bind HER2 antigens.
- Bound HER2 changes the local electrostatic environment near the graphene channel.
- This perturbation changes the drain-source current, ΔI_{DS} .
- The signal is modeled analytically from bound molecule count and coupling efficiency.

$$\Delta I_{DS} = \frac{W}{L} e \mu V_{DS} \frac{\alpha N}{A_{eff}}$$

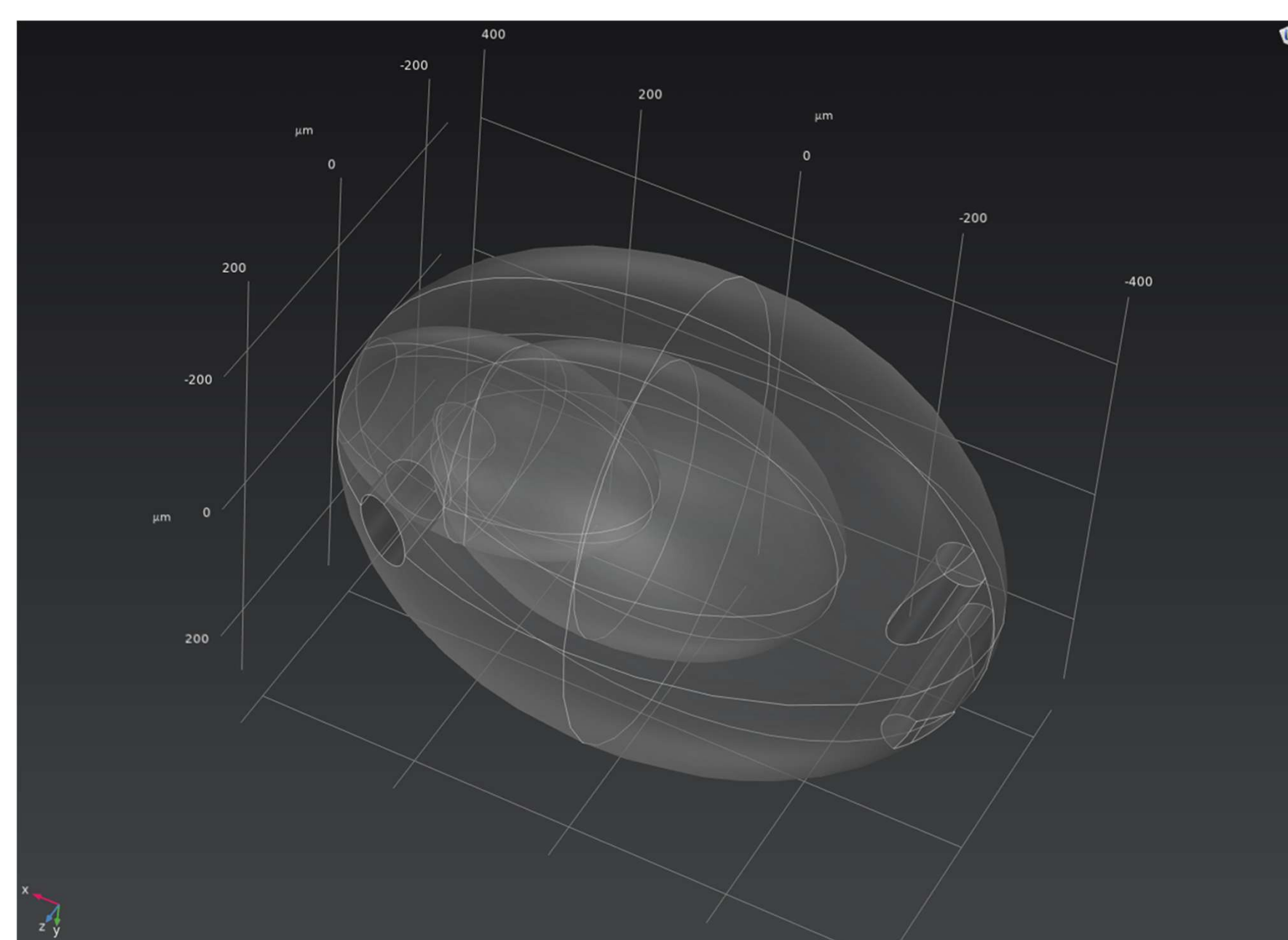
HER2 biosensor flow: Transport to Design Requirements



Simulation Pipeline

- M1: HER2-antibody binding kinetics.
- M2: diffusion-only COMSOL transport baseline.
- M2B: partially anatomical prescribed-flow convection-diffusion extension.
- M3/M4: surface binding, bound molecule count, GFET current, LOD.
- Experiments C-E: detectability, Debye screening, and design requirements.

M2B is a partially anatomical prescribed-velocity extension, not a full anatomical lymph-node model.

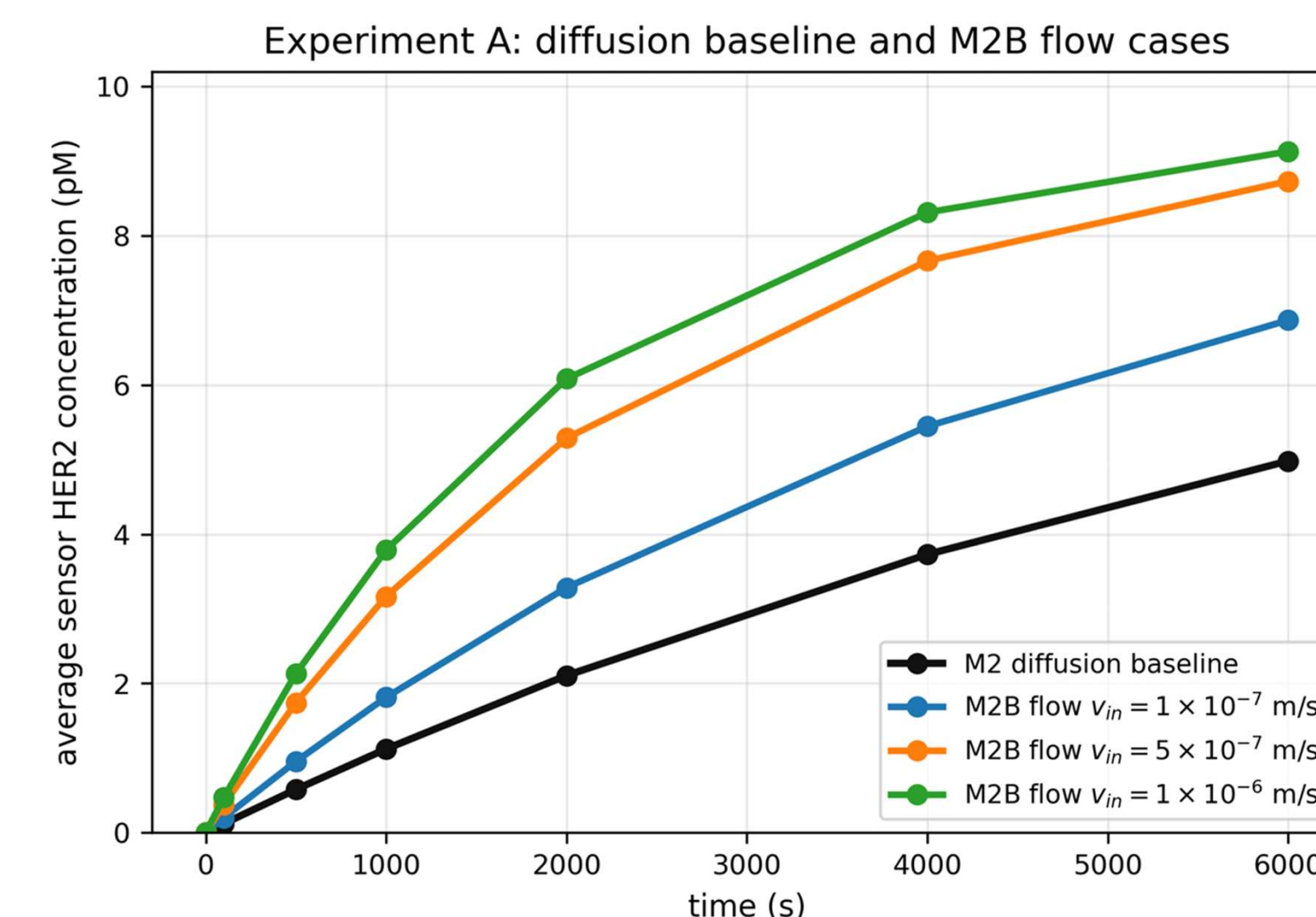


Lymph-Node-Inspired Transport Model

- Outer region approximates cortex/capsule.
- Inner region approximates medulla.
- M2 baseline isolates heterogeneous diffusion.
- M2B adds directional prescribed flow with afferent inlet regions and a hilum-side outlet.
- The goal is to estimate HER2 exposure at the GFET sensing region.

Convection-diffusion transport in M2B:

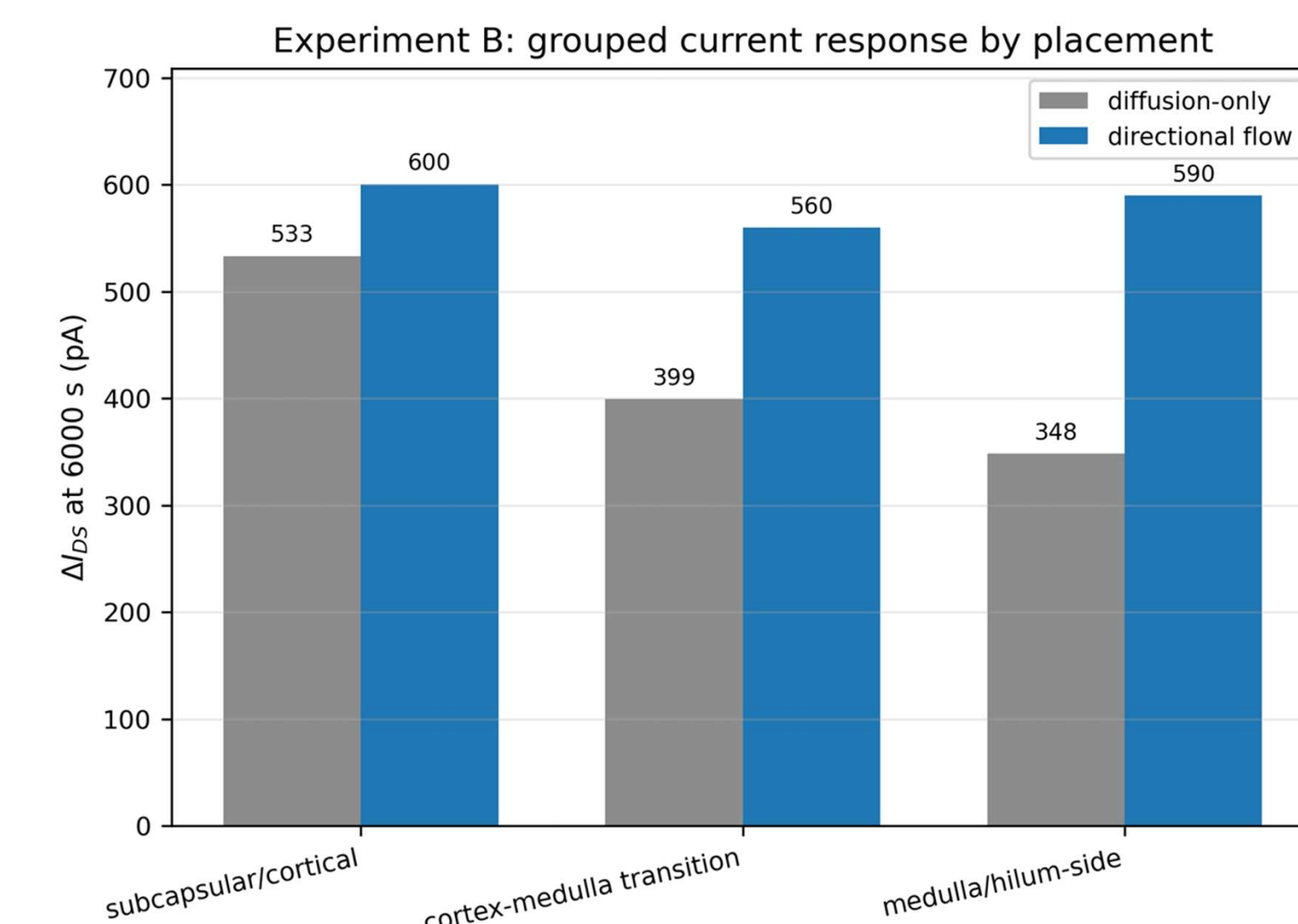
$$\frac{\partial c}{\partial t} + \mathbf{u} \cdot \nabla c = \nabla \cdot (D \nabla c)$$



Result 1 - Directional Flow Improves Exposure

- At 10 pM HER2, diffusion-only transport gives 4.98 pM sensor concentration at 6000 s.
- Reference flow, $v_{in} = 5 \times 10^{-7}$ m/s, gives 8.73 pM at 6000 s.
- This corresponds to approximately 1.75x higher late-time exposure.
- Time to 50% exposure decreases from 6038 s to 1839 s.

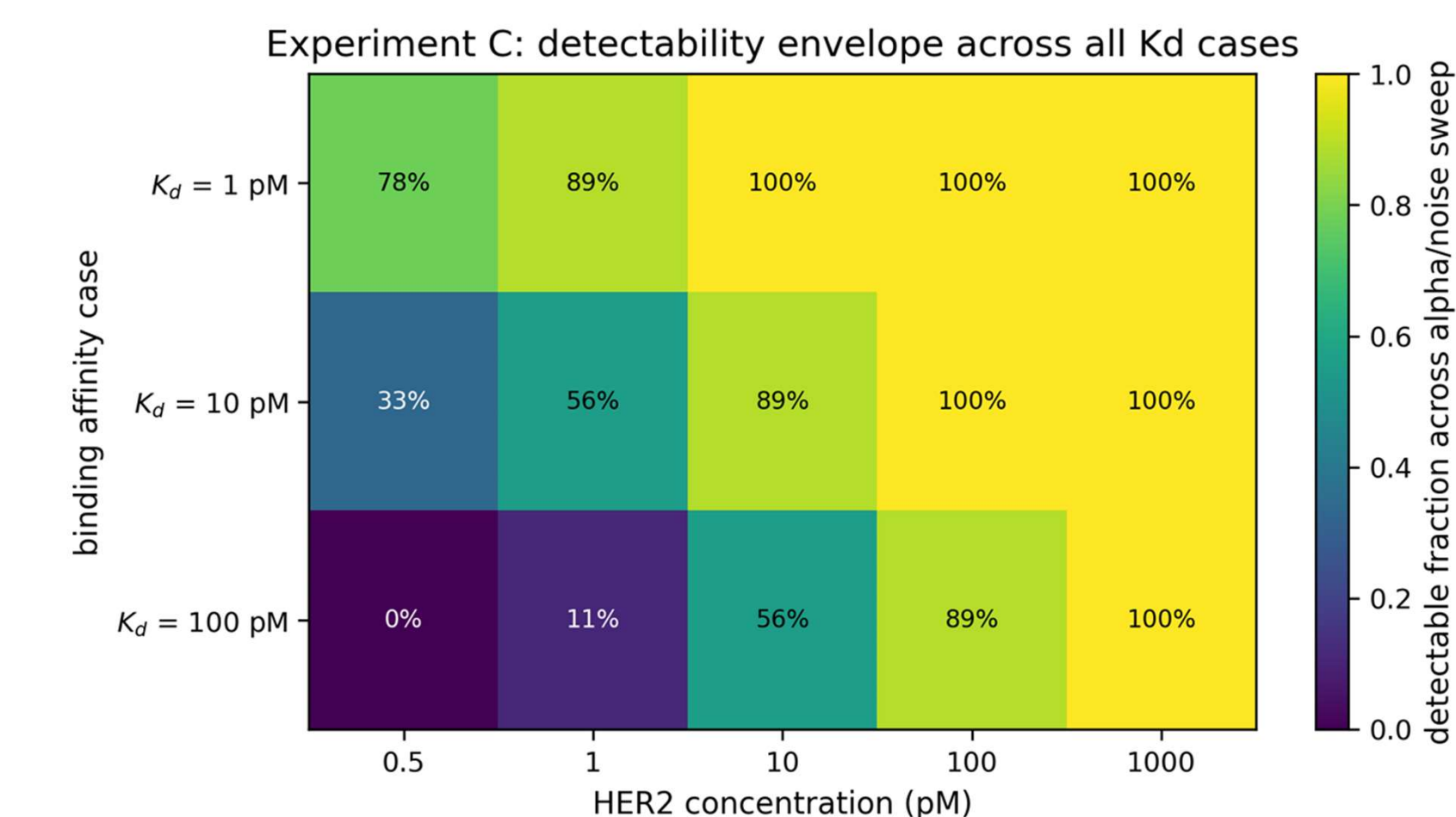
Flow improves exposure, but transport is only the first bottleneck.



Result 2 - Sensor Placement Matters

- Three sensor regions were evaluated: subcapsular/cortical, cortex-medulla transition, and medulla/hilum-side.
- Subcapsular/cortical placement gives the strongest response under reference flow.
- Current response at 6000 s increases under directional flow for all tested placements.
- Placement affects response timing and pathway exposure.

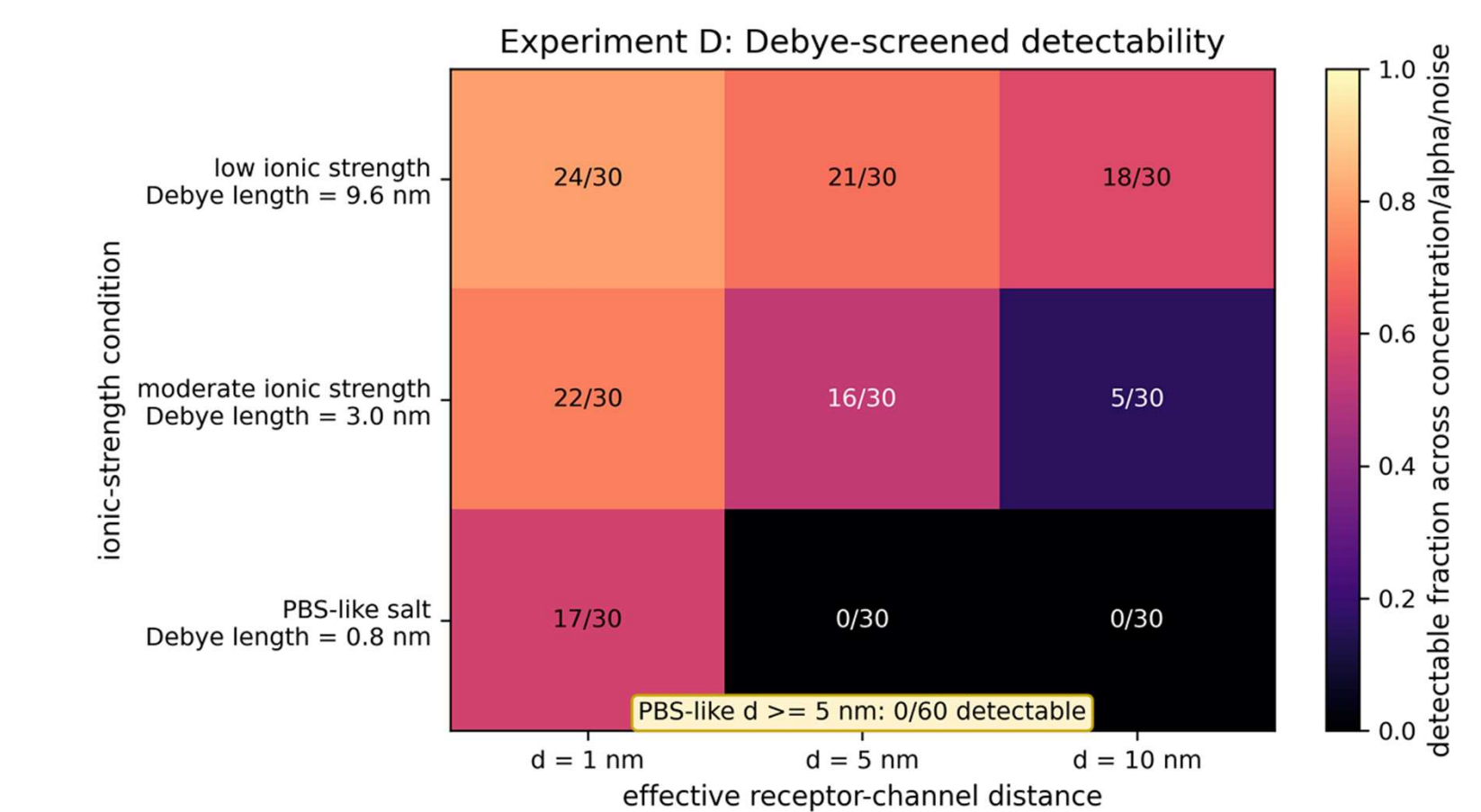
Placement sweep is post-processed over the transport model, not a full remeshed COMSOL relocation study.



Result 3 - Detection Is Not Universal

- Detectability was tested across HER2 concentration, K_D , coupling efficiency, and noise floor.
- 99 parameter combinations passed the 3σ detection threshold.
- 36 parameter combinations failed.
- Detection improves with higher concentration, tighter binding, stronger coupling, and lower noise.

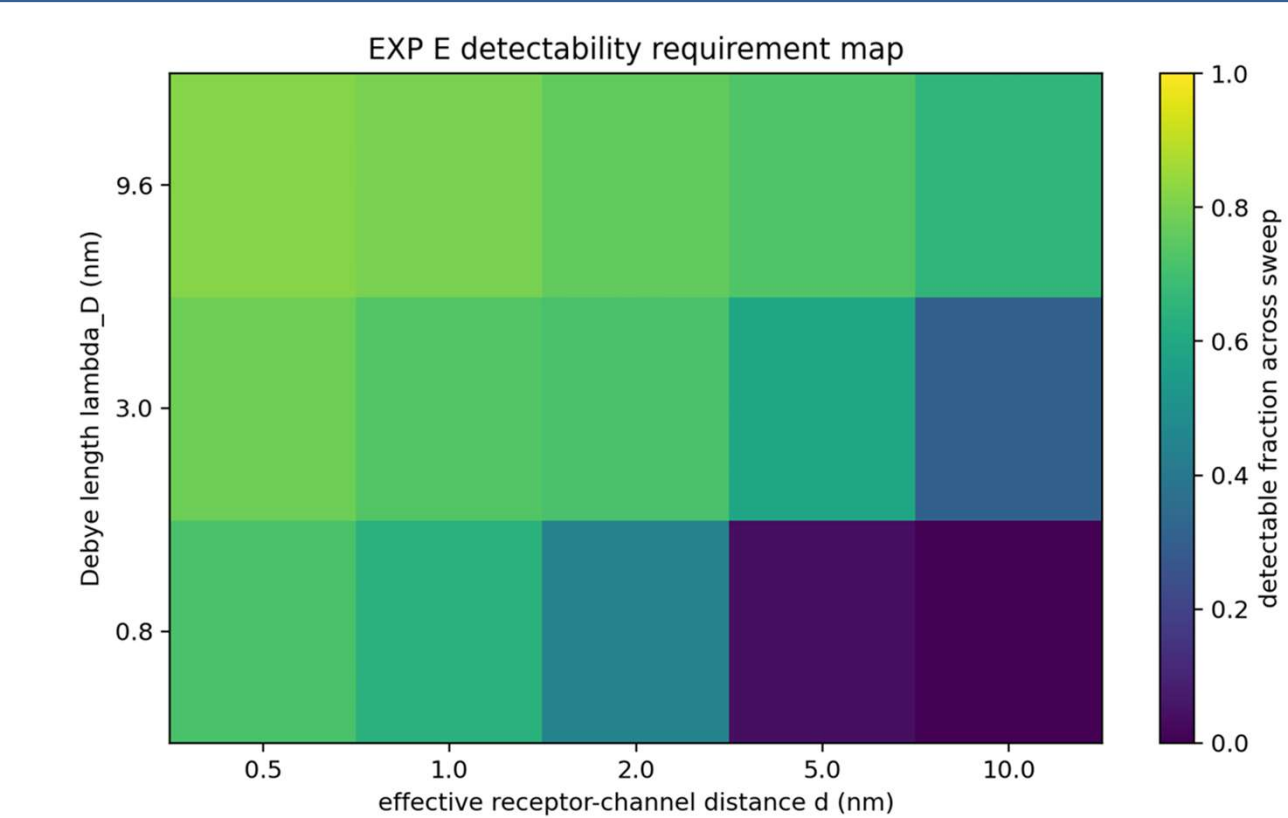
Sub-pM detection is possible only under favorable assumptions.



Result 4 - Physiological Screening Suppresses the Signal

- Debye screening attenuates effective GFET coupling: $\alpha_{eff} = \alpha_0 \exp(-d / \lambda_D)$
- PBS-like screening uses $\lambda_D \approx 0.8$ nm.
- For PBS-like screening and $d \geq 5$ nm, 0/60 screened cases remain detectable.
- Transport improvement alone cannot overcome electrostatic screening.

Main negative result: even if HER2 reaches the sensor, the signal can be screened out.



Result 5 - Strict Design Conditions Are Required

- Experiment E converts screening failure into design requirements.
- Under PBS-like screening, detection is not robust beyond ~1 nm effective receptor-channel distance for the baseline case.
- For $K_D = 10$ pM, 10 pA noise, PBS-like screening, $d = 1$ nm: required $\alpha_0 \approx 0.0326$ at 1 pM HER2, required $\alpha_0 \approx 0.00561$ at 10 pM HER2
- Feasible detection requires short receptor distance, strong coupling, low noise, and screening control

Output: design requirements, not clinical proof.

Key Parameters

- HER2 concentration: 0.5–1000 pM
- Baseline C: 10 pM
- K_D sweep: 1, 10, 100 pM
- M2B velocity: 0 to 1×10^{-6} m/s
- α_0 sweep: 0.01, 0.03, 0.1
- Noise floor: 1–100 pA
- Debye length: 9.6, 3.0, 0.8 nm
- Receptor-channel distance: 0.5–10 nm

Main Engineering Conclusion

- Directional flow increases HER2 exposure at the sensing region.
- Sensor placement affects response timing and current response.
- Detectability is conditional, not guaranteed.
- Under physiological-like screening, receptor-channel distance and electronics noise dominate feasibility.
- A feasible GFET-HER2 design requires short receptor distance, strong coupling, low noise, favorable binding, and screening mitigation.

Transport can be improved; electrical detectability remains the hard constraint.

Limitations & Future Work

- Simulation-only study;** no fabricated GFET or clinical validation.
- M2B uses **prescribed velocity**, not validated Darcy/Brinkman lymph flow.
- GFET response and Debye screening are **analytical feasibility models**.
- Future work:** experimental binding validation, low-noise readout, shorter receptor-channel distance, and improved flow/electrostatic modeling.

